

ECG Analyzer: The Anatomy of a Heartbeat

A Real-Time Cardiac Diagnostics Pipeline

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Abstract

This document follows a single strip of raw electrocardiogram (ECG) voltage as it travels through a complete real-time cardiac diagnostics pipeline — from a noisy analog-like trace pulled off the PhysioNet MIT-BIH Arrhythmia Database, through denoising, QRS detection via the Pan–Tompkins algorithm, heart-rate estimation, time- and frequency-domain Heart Rate Variability (HRV) analysis, beat-morphology classification, and finally into a rule-based rhythm diagnosis. Each stage is presented as a chapter in the signal’s transformation: what problem it solves, the mathematics behind it, and why that particular choice was necessary before the next stage could begin.

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Part I

Project Documentation

1 Motivation behind choosing this project

Our life begins with a signal — the very first heartbeat inside the womb of our mother. Whatever paths our lives may take, this signal continues faithfully until our departure from this world. To us, the electrocardiogram is not merely a biomedical trace; it is a bridge between our material studies and the essence of human existence. The heart is the most important organ in our body: if the heart is sound, the whole body is sound; if the heart is corrupted, the whole body is corrupted. This is not just a feeling, but a truth mentioned in the hadith: **“Verily, in the body there is a piece of flesh which, if it is sound, the whole body is sound; and if it is corrupt, the whole body is corrupt. Verily, it is the heart.”*

Working around this profound signal fills our project with joy and enthusiasm. For us in Team Signum, choosing this project was not simply an academic decision, but a way of listening to the whispers of life itself. We tried to catch and interpret the signals that the heart is always giving us, and our newly added module of cardiac diagnostics stands entirely on this motivation: that learning the signals within our own body is among the most important things we can do. We both loved our project, and through it we felt connected not only to science, but to the truth of our own humanity.

2 Problem Description

Electrocardiogram (ECG) signals are one of the most widely used physiological measurements for monitoring heart health, but raw ECG recordings captured by real sensors and hospital equipment are rarely clean. They are typically corrupted by baseline wander (from patient breathing and body movement), 50/60 Hz powerline interference, and muscle-artifact noise. This corruption makes both manual visual inspection and automated downstream analysis — such as heartbeat detection — unreliable if the raw signal is used directly.

This project addresses the problem of extracting reliable, clinically meaningful information from noisy, real-world ECG recordings using classical signal processing techniques, rather than opaque machine learning models. The intended users are students and instructors studying biomedical signal processing, and more broadly, this pipeline mirrors the front-end processing stage used in real cardiac monitoring devices and wearables before any diagnostic decision is made.

3 Project Evolution: From Detection to Diagnosis

The scope of this project changed over the course of its development, and that change of direction is itself worth documenting clearly, since it explains why the system contains more than what was originally proposed.

3.1 Version 1: R-Peak Detection and Validation

The project was originally scoped around a single, focused question: *can a classical DSP pipeline reliably find every heartbeat in a noisy, real-world ECG recording, and how accurate is it against expert ground truth?* This version consisted of exactly two functional modules:

1. **Preprocessing and QRS Detection** — notch and bandpass filtering followed by the Pan–Tompkins algorithm to locate R peaks.

2. **Validation** — comparing detected R peaks against the MIT-BIH database’s expert-annotated ground truth to report precision and recall.

Heart-rate estimation and FFT-based spectral visualization of the raw versus filtered signal supported this core detection-and-validation loop, but no further clinical interpretation was attempted.

3.2 Version 2: The Cardiac Diagnostics Module

Following supervisor feedback, the project’s scope was deliberately expanded beyond “finding beats” toward “interpreting what the beats mean.” A new **Cardiac Diagnostics module** was introduced, adding four capabilities that did not exist in the original proposal:

1. **Time-Domain HRV Analysis** — SDNN, RMSSD, and pNN50, computed directly from the R-R interval series.
2. **Frequency-Domain HRV Analysis** — uniform resampling of the R-R series, a from-scratch DFT/periodogram implementation, Welch’s method for PSD estimation, and LF/HF band-power ratio computation.
3. **Beat Morphology Analysis** — template construction from averaged beats and cosine-similarity-based flagging of abnormally shaped beats.
4. **Rule-Based Rhythm Classification** — a decision layer that combines heart rate, HRV metrics, and abnormal-beat counts into tachycardia, bradycardia, irregular-rhythm, and abnormal-beat flags, culminating in a generated diagnostic report.

This is reflected in the current API surface as a dedicated `/diagnose/{record_name}` endpoint, sitting alongside the original `/analyze/{record_name}` and `/validate/{record_name}` endpoints rather than replacing them — so the original detection-and-validation pipeline remains intact and independently testable, with diagnostics layered on top of it.

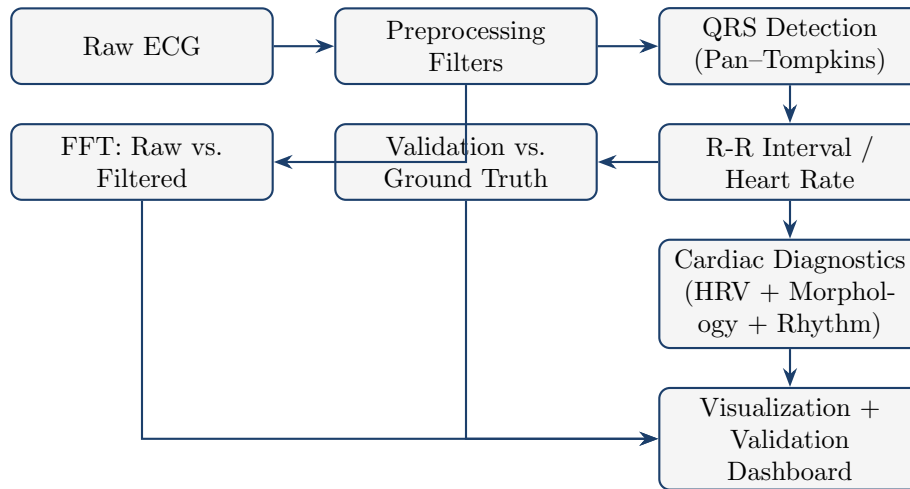
4 Proposed Solution

Input: Real, publicly available ECG recordings from the PhysioNet MIT-BIH Arrhythmia Database, which additionally provides expert cardiologist-annotated ground-truth R-peak locations for every recording, allowing detection accuracy to be validated quantitatively rather than only inspected visually.

Processing Pipeline (current, post-expansion):

1. **Preprocessing** — a notch filter removes 50/60 Hz powerline interference, and a bandpass filter removes baseline wander and high-frequency noise.
2. **QRS (Heartbeat) Detection** — the classical Pan–Tompkins pipeline: bandpass filtering, derivative computation, squaring, moving-window integration, and adaptive thresholding.
3. **Frequency-Domain Analysis** — FFT computed on both raw and filtered signals to visualize noise removal and justify filter cutoff choices.
4. **Heart Rate Estimation** — computed from R-R intervals.
5. **Validation** — detected R peaks compared against ground-truth annotations to report precision and recall.
6. **Cardiac Diagnostics** (added module) — time- and frequency-domain HRV analysis, beat-morphology anomaly detection, and rule-based rhythm classification, culminating in a generated diagnostic report.

Output: A cleaned ECG waveform, an annotated waveform with detected heartbeats marked, a heart-rate readout, a frequency spectrum view, a detection-accuracy report, and — from the added module — an HRV metrics summary, a set of flagged abnormal beats, and a rule-based rhythm classification with diagnostic notes.



5 Signal Processing Concepts Used

- **Convolution and LTI Systems** — all filtering stages (baseline-wander removal, notch filtering, and the Pan-Tompkins derivative and moving-window integration steps) are implemented as convolution with a designed impulse response, directly applying LTI system theory.
- **Filter Design (FIR/IIR)** — bandpass and notch filters are designed and their frequency response justifies cutoff selection, applying frequency response and pole-zero concepts.
- **Fourier Transform (FFT/DFT)** — used both to inspect raw-versus-filtered spectral content and, in the diagnostics module, to build the Welch PSD estimate underlying LF/HF analysis.
- **Sampling and Discrete-Time Signals** — ECG recordings are discrete-time sampled signals; R-R interval timing, heart-rate estimation, and RR-series resampling all rely on correct handling of the sampling rate.
- **Thresholding / Decision Logic on Processed Signals** — adaptive thresholding on the integrated signal to localize heartbeats, and cosine-similarity thresholding to flag abnormal beat morphology.
- **Power Spectral Density** — originally listed as a stretch goal, this became a core part of the Cardiac Diagnostics module, used for frequency-domain HRV (LF/HF) analysis.

6 User Interface Design

The interface allows a user to select or upload an ECG recording, then view its analysis through several linked, synchronized panels:

1. **Raw Signal Panel** — the unprocessed ECG waveform as recorded.
2. **Filtered Signal Panel** — the cleaned waveform after baseline-wander and powerline-noise removal.
3. **Heartbeat Detection Panel** — the filtered waveform with detected R peaks marked.
4. **Frequency Spectrum Panel** — side-by-side FFT magnitude plots of the raw versus filtered signal.
5. **Validation Dashboard** — precision/recall of detected heartbeats against expert-annotated ground truth for the currently loaded recording.
6. **Cardiac Diagnostics Panel** (added module) — HRV metrics (SDNN, RMSSD, pNN50, LF/HF), the PSD plot, flagged abnormal beats overlaid on the waveform, and the rule-based rhythm classification with report notes.

The layout uses a tabbed, multi-panel dashboard so a user can move between recordings and immediately see how each processing stage transforms the signal, from raw voltage through to a final diagnostic verdict.

7 Project Justification

This project is directly built on core signal-processing topics: convolution and LTI system behavior, Fourier analysis, sampling theory, and filter design. Rather than treating these as isolated exercises, the project applies them together in a single, coherent real-world pipeline, from raw noisy sensor data to a validated clinical measurement — and, after the scope expansion, onward to a rule-based clinical interpretation layer.

The extended timeline reflects the project's several distinct, non-trivial stages, each implemented, tested, and integrated with a live UI: filter design and validation, the multi-stage Pan–Tompkins detection algorithm, FFT-based spectral visualization, quantitative accuracy validation against ground-truth annotations, the added time-/frequency-domain HRV and beat-morphology analysis, rule-based rhythm classification, and a responsive multi-panel frontend with interactive controls. Because real ECG data is used rather than synthetic signals, meaningful time was also needed for testing across multiple recordings with different noise characteristics and arrhythmia types.

8 Tools and Technologies

Backend: Python, with NumPy and SciPy for filter design, convolution, and FFT computation, and the `wfdb` package to read ECG records and annotations directly from the PhysioNet MIT-BIH Arrhythmia Database. A FastAPI server exposes processed results — including the added diagnostics endpoint — to the frontend.

Frontend: React, with a charting library (e.g. Plotly.js or Recharts) for interactive, linked waveform and spectrum visualizations, and slider controls for live filter-parameter adjustment.

Data: PhysioNet MIT-BIH Arrhythmia Database — real, annotated ECG recordings, freely available for academic use.

9 Github Repository Link

You can find our project repository here:

<https://github.com/SumaiyaSultanaLabiba/ecg-analyzer>

10 Live Demonstration Link on Render

The live demo of our ECG Analyzer pipeline is hosted on Render:

<https://electrocradiogram-analyzer.onrender.com>

Part II

The Pipeline: A Narrative Walkthrough

11 Prologue: A Signal is Born

Every heartbeat begins as an electrical event. Cardiac muscle cells depolarize and repolarize in a coordinated wave, and electrodes placed on the skin pick up the resulting potential difference as a continuous voltage $x(t)$. When this analog signal is digitized at a sampling frequency f_s , we obtain a discrete sequence

$$x[n] = x(nT_s), \quad T_s = \frac{1}{f_s}, \quad n = 0, 1, 2, \dots, N-1.$$

In this pipeline, $x[n]$ is drawn from the **MIT-BIH Arrhythmia Database** (`mitdb`) via the `wfdb` library, giving us both the raw signal and its ground-truth expert-annotated beat locations — the latter serving, later in our story, as the yardstick against which our own detector is judged.

But raw $x[n]$ is not yet trustworthy. It is contaminated: by the 50/60 Hz hum of nearby electrical equipment, by slow baseline wander from breathing and electrode drift, and by muscle-noise artifacts. Before any diagnosis can be attempted, the signal must first be *cleaned*.

12 Chapter I: Denoising — Finding the Signal Beneath the Noise

12.1 The Notch Filter: Silencing the Hum

Powerline interference appears as a narrow, almost pure sinusoidal component at a fixed frequency f_0 (60 Hz in this pipeline). An **IIR notch filter** is designed to surgically remove only this frequency while leaving its neighbors untouched. Its transfer function has the general biquad form

$$H(z) = \frac{1 - 2\cos(\omega_0)z^{-1} + z^{-2}}{1 - 2r\cos(\omega_0)z^{-1} + r^2z^{-2}}, \quad \omega_0 = \frac{2\pi f_0}{f_s},$$

where the quality factor Q controls how narrow the rejected band is ($Q = 30$ here means a very sharp, surgical notch — wide enough to catch the hum, narrow enough to spare the ECG's own content near that frequency).

12.2 The Bandpass Filter: Keeping Only What Matters

A healthy ECG's diagnostically useful energy lives roughly between 0.5 Hz and 40 Hz. Below this, baseline wander dominates; above it, muscle noise and quantization artifacts take over. A **Butterworth bandpass filter** of order k is applied:

$$|H(j\Omega)|^2 = \frac{1}{1 + \left(\frac{\Omega}{\Omega_c}\right)^{2k}},$$

chosen for its *maximally flat passband* — it does not ripple the frequencies we want to keep, at the cost of a gentler roll-off outside the band. The cutoff frequencies are normalized against the Nyquist frequency $f_s/2$:

$$\omega_{low} = \frac{f_{low}}{f_s/2}, \quad \omega_{high} = \frac{f_{high}}{f_s/2}.$$

12.3 Zero-Phase Filtering: Undoing Time Itself

A single causal IIR filter pass introduces a *phase delay* — it shifts features in time. Since our very next task is to *locate* peaks precisely, any such shift would corrupt every downstream timing measurement. The pipeline instead applies `scipy.signal.filtfilt`, which filters the signal once forward and once *backward*:

$$y[n] = (h * (h_{rev} * x))[n],$$

so that the phase distortions introduced by each pass exactly cancel, at the cost of doubling the effective filter order. The result, $x_{clean}[n]$, is a signal with the same timing as the original but with noise outside the diagnostic band removed — our first transformed character in this story.

13 Chapter II: The Pan–Tompkins Algorithm — Finding the Heart’s Pulse

With a clean signal in hand, the next task is deceptively simple to state and surprisingly subtle to solve: *find every R peak* — the sharp spike marking each heartbeat. This pipeline implements the classic **Pan–Tompkins algorithm** (Pan & Tompkins, 1985), a five-act transformation designed to make R peaks impossible to miss and everything else fade into the background.

13.1 Act 1 — A Second Bandpass: Isolating the QRS

A tighter bandpass, 5–15 Hz, is applied to $x_{clean}[n]$. The QRS complex concentrates most of its spectral energy in exactly this band, so this step suppresses the P wave, T wave, and any remaining noise, leaving mostly the QRS shape:

$$x_{qrs}[n] = \text{BPF}_{5-15\text{Hz}}(x_{clean}[n]).$$

13.2 Act 2 — Derivative: Chasing the Steepest Slope

The QRS complex is defined by its abrupt rise and fall — the steepest slope anywhere in the cardiac cycle. A five-point derivative approximation estimates this slope:

$$y[n] = \frac{f_s}{8} (2x[n] + x[n-1] - x[n-3] - 2x[n-4]).$$

Flat regions of the signal (P wave, T wave, baseline) produce small derivative values; the QRS complex produces a large one. This is a discrete approximation of $\frac{dx}{dt}$, weighted to reduce high-frequency noise sensitivity while remaining responsive to genuine slope changes.

13.3 Act 3 — Squaring: Turning Slope into Energy

$$z[n] = y[n]^2.$$

Squaring serves two purposes at once: it makes every value non-negative (so positive and negative slopes reinforce rather than cancel when summed later), and it is *nonlinear* — large derivative values are amplified disproportionately more than small ones, sharply increasing the contrast between the QRS complex and background noise.

13.4 Act 4 — Moving Window Integration: Smoothing Into a Hump

$$w[n] = \frac{1}{N_w} \sum_{k=0}^{N_w-1} z[n-k], \quad N_w = \lfloor 0.150 f_s \rfloor,$$

a moving average over a 150 ms window. This converts the sharp, jagged squared-derivative signal into one smooth “hump” of energy per heartbeat, whose *width* approximates the duration of the QRS complex itself and whose *peak* sits close to (but not exactly at) the true R peak.

13.5 Act 5 — Adaptive Thresholding and Refinement

Candidate peaks are located on $w[n]$ subject to two physiologically motivated constraints:

$$\text{threshold} = 1.5 \cdot \overline{w[n]}, \quad \text{min. distance} = \lfloor 0.200 f_s \rfloor,$$

the latter encoding the simple fact that the human heart cannot exceed roughly 300 beats per minute, so no two true R peaks can be closer than 200 ms.

However, $w[n]$ is a heavily smoothed and *time-shifted* representation — its peak location is not the true R peak location in the original signal. A final refinement step searches a small ± 75 ms window around each candidate, directly in $x_{\text{clean}}[n]$, and snaps to the true local maximum:

$$\hat{r}_i = \arg \max_{n \in [c_i - w_s, c_i + w_s]} x_{\text{clean}}[n], \quad w_s = \lfloor 0.075 f_s \rfloor,$$

where c_i is the i -th candidate index. The set $\{\hat{r}_i\}$ is the pipeline’s final answer to the question “where did each heartbeat occur?”

Stage	Transformation	Purpose
1	Bandpass 5–15 Hz	Isolate QRS frequency content
2	5-point derivative	Emphasize steep slopes
3	Squaring	Positivity + nonlinear amplification
4	Moving window integration	Smooth into one hump per beat
5	Threshold + refine	Locate true R peak, enforce physiology

14 Chapter III: Heart Rate — The Simplest Question

Once R peaks $\hat{r}_1, \hat{r}_2, \dots, \hat{r}_M$ are known, the instantaneous interval between consecutive beats is

$$\text{RR}_i = \frac{\hat{r}_{i+1} - \hat{r}_i}{f_s} \quad (\text{seconds}),$$

and the mean heart rate follows directly from the mean interval:

$$\text{HR}_{\text{bpm}} = \frac{60}{\overline{\text{RR}}}.$$

This single number answers the first clinical question — *is the heart beating too fast, too slow, or normally?* — but it discards the far richer story hidden in *how* that interval varies from beat to beat. That story is the subject of Heart Rate Variability analysis.

15 Chapter IV: Heart Rate Variability in Time — The Rhythm’s Fingerprint

Definition 1. The **RR interval series** $\{\text{RR}_i\}_{i=1}^{M-1}$ is the sequence of beat-to-beat time intervals. Unlike a directly sampled signal, its samples are not evenly spaced in time — each RR value occurs at the moment of its corresponding beat.

Three classical time-domain metrics summarize this variability:

15.1 SDNN — Overall Variability

$$\text{SDNN} = \sqrt{\frac{1}{M-2} \sum_{i=1}^{M-1} (\text{RR}_i - \overline{\text{RR}})^2}$$

the sample standard deviation (ddof = 1) of all RR intervals — the broadest, simplest measure of overall variability, sensitive to both short- and long-term fluctuation.

15.2 RMSSD — Short-Term, Beat-to-Beat Variability

$$\text{RMSSD} = \sqrt{\frac{1}{M-2} \sum_{i=1}^{M-2} (\text{RR}_{i+1} - \text{RR}_i)^2}$$

the root-mean-square of *successive differences*. Because it looks only at consecutive beats, RMSSD is particularly sensitive to fast changes and is widely treated as a proxy for parasympathetic (“rest and digest”) nervous system activity.

15.3 pNN50 — A Threshold’s-Eye View

$$\text{pNN50} = \frac{\#\{i : |\text{RR}_{i+1} - \text{RR}_i| > 0.05\text{s}\}}{M-2} \times 100\%$$

the percentage of successive differences exceeding 50 ms — a coarser, threshold-based cousin of RMSSD that asks not *how much* the rhythm varies, but *how often* it varies by a clinically meaningful amount.

16 Chapter V: Heart Rate Variability in Frequency — The Rhythm’s Hidden Song

16.1 The Problem of Uneven Time

The RR series lives on an irregular timeline: each value is timestamped at its own beat, and beats do not occur at a constant rate. Classical spectral tools (the DFT, FFT, Welch’s method) all assume *uniformly sampled* input. Before any frequency analysis can proceed, the series must first be resampled onto an evenly spaced time grid via linear interpolation:

$$\text{RR}_{\text{unif}}[m] = \text{interp}(t_m; \{t_i^{\text{RR}}\}, \{\text{RR}_i\}), \quad t_m = t_0 + \frac{m}{f_{rs}}, \quad m = 0, 1, 2, \dots$$

Remark 1 (Why $f_{rs} = 4\text{ Hz}$?). The resampling rate is not arbitrary; it satisfies three simultaneous constraints:

1. **Nyquist:** the highest frequency of clinical interest (the top of the HF band, 0.4 Hz) requires $f_{rs} \geq 0.8\text{ Hz}$. 4 Hz gives five-fold headroom.
2. **Natural beat rate:** even a fast heart (180 bpm $\approx 3\text{ Hz}$) produces beats slower than 4 Hz, so resampling upward loses no information already present.
3. **Convention:** 4 Hz is the standard adopted by the Task Force of the European Society of Cardiology / NASPE guidelines on HRV measurement, making results directly comparable across the literature.

16.2 The Discrete Fourier Transform

At the mathematical heart of this chapter lies the DFT itself, implemented here from first principles rather than via a fast algorithm:

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j2\pi kn/N}, \quad k = 0, 1, \dots, N-1,$$

a direct $O(N^2)$ evaluation of the transform — pedagogically transparent, though FFT would compute the identical result in $O(N \log N)$.

16.3 The Windowed Periodogram

Before transforming, each segment is tapered by a **Hamming window** $h[n]$ to suppress *spectral leakage* — the smearing of energy across frequencies caused by an abrupt segment boundary. Because windowing removes energy, a normalization factor

$$U = \frac{1}{N} \sum_{n=0}^{N-1} h[n]^2$$

restores the correct power scale. The one-sided Power Spectral Density estimate is then

$$\hat{S}_{xx}[k] = \frac{|X[k]|^2}{f_s N U}, \quad k = 0, \dots, \left\lfloor \frac{N}{2} \right\rfloor,$$

with all non-edge bins doubled to account for the discarded (mirrored) negative-frequency half of a real-valued signal's spectrum.

16.4 Welch's Method: Averaging Away the Noise

A single periodogram is a noisy, high-variance estimator. **Welch's method** instead splits the signal into L overlapping segments of length $N_{seg} = 256$ (with 50% overlap, $N_{ov} = 128$), computes a periodogram $\hat{S}_{xx}^{(\ell)}[k]$ for each, and averages:

$$\hat{S}_{xx}^{Welch}[k] = \frac{1}{L} \sum_{\ell=1}^L \hat{S}_{xx}^{(\ell)}[k].$$

This trades a small amount of frequency resolution for a substantial reduction in estimator variance — the classical bias–variance tradeoff of spectral estimation.

16.5 LF and HF Bands: Reading the Autonomic Nervous System

The resulting spectrum is integrated (via the trapezoidal rule) over two bands established by decades of HRV research:

$$P_{LF} = \int_{0.04}^{0.15} S_{xx}(f) df, \quad P_{HF} = \int_{0.15}^{0.40} S_{xx}(f) df,$$

and their ratio,

$$\text{LF/HF} = \frac{P_{LF}}{P_{HF}},$$

is interpreted — with appropriate caution, as this ratio is debated in the modern HRV literature — as a rough marker of sympathetic/parasympathetic balance. HF power in particular tracks closely with respiration and parasympathetic (vagal) tone.

17 Chapter VI: The Shape of a Beat — Morphology and Anomaly

Timing alone cannot catch every abnormality; a beat can arrive exactly on schedule and still have the wrong *shape*. This chapter shifts from *when* to *what*.

17.1 Building a Template

Around every R peak $\hat{r}_i$, a fixed window is extracted from the *clean* signal $x_{clean}[n]$ (not the QRS-only bandpassed signal — that would have already discarded the P and T waves needed to judge true morphology):

$$\beta_i = x_{clean}[\hat{r}_i - n_{pre} : \hat{r}_i + n_{post}], \quad n_{pre} = 0.2f_s, \quad n_{post} = 0.4f_s.$$

Averaging across all valid beats produces a **template** τ — the “typical” heartbeat shape for this recording:

$$\tau = \frac{1}{M} \sum_{i=1}^M \beta_i.$$

17.2 Cosine Similarity: Measuring Shape, Not Size

Each beat is then compared against τ using the cosine similarity:

$$\rho_i = \frac{\beta_i \cdot \tau}{\|\beta_i\| \|\tau\|} = \frac{\sum_n \beta_i[n] \tau[n]}{\sqrt{\sum_n \beta_i[n]^2} \sqrt{\sum_n \tau[n]^2}} \in [-1, 1].$$

This measures the *angle* between two beats treated as vectors — invariant to amplitude scaling, sensitive only to shape. A beat with $\rho_i < 0.8$ is flagged as morphologically abnormal — a possible ectopic beat, artifact, or arrhythmic event.

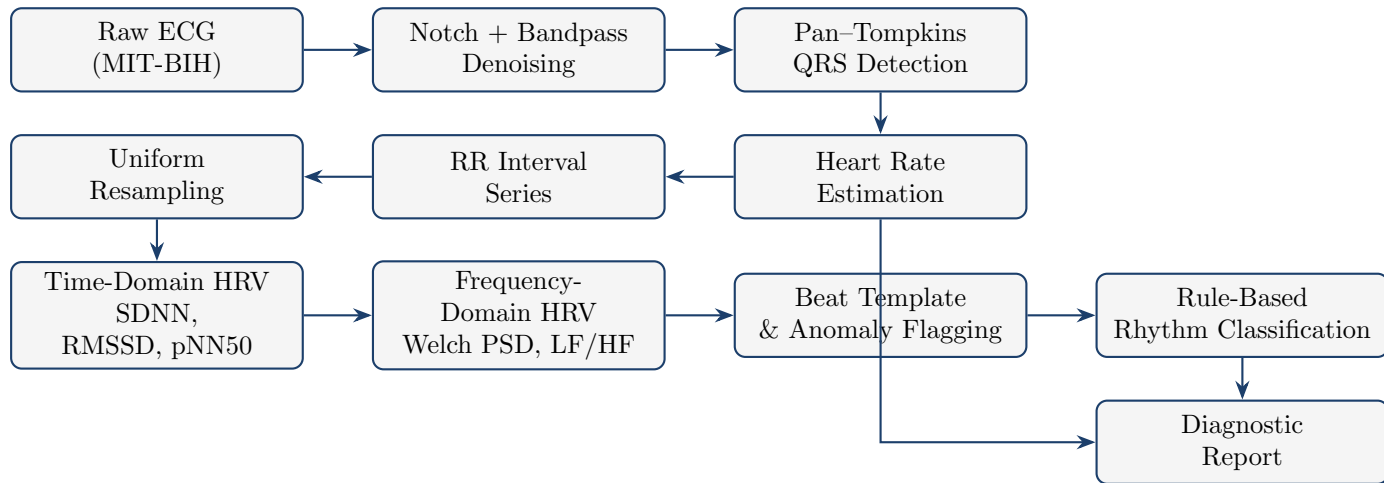
18 Chapter VII: Judgment — From Numbers to Diagnosis

The final act gathers every earlier measurement into a simple rule-based classifier:

$$\text{flags} = \begin{cases} \text{tachycardia}, & \text{HR}_{bpm} > 100 \\ \text{bradycardia}, & \text{HR}_{bpm} < 60 \\ \text{irregular_rhythm}, & \text{SDNN} > 100\text{ms} \\ \text{abnormal_beats_detected}, & N_{abn} > 0 \end{cases}$$

Each condition is independently evaluated; the presence of *any* flag marks the overall recording status as **abnormal**, otherwise **normal**. This is the pipeline’s final verdict — reached not through a single measurement, but through the accumulated evidence of everything that came before.

19 Epilogue: The Full Pipeline



What began as an anonymous column of voltage readings, indistinguishable from noise, has — through denoising, transformation, thresholding, integration, and rule-based reasoning — become a structured clinical judgment: a heart rate, a variability profile, a set of flagged beats, and a final verdict. Every equation in this document exists for exactly one reason: to extract a little more truth from the same ten seconds of signal.

“Love your heart; listen to what it signals.”

—End of document—